

TSA6213 SYSTEM ADMINISTRATION & MAINTENANCE

LAB 3

Name	Muhammad Eusoff Aminurrashid Bin Shukri
Student ID	1211112228
Section	SAM1, 1A

1. Package Management – dpkg and apt [advance package tool] Commands

a) List and count ALL installed packages. Use dpkg with suitable option and pipe it to nl or wc command to count the packages.

```
dpkg --get-selections | wc -l
```

[count all installed packages]

```
osboxes@osboxes:~$ dpkg --get-selections | wc -l
1524
osboxes@osboxes:~$
```

```
dpkg --get-selections | nl
```

[list all installed packages with line numbers]

```
osboxes@osboxes:~$ dpkg --get-selections | nl
 1 accountsservice          install
 2 acl                      install
 3 acpi-support             install
 4 acpid                   install
 5 adduser                 install
 6 adwaita-icon-theme      install
 7 alsa-base               install
 8 alsa-topology-conf      install
 9 alsa-ucm-conf           install
10 alsa-utils              install
11 amd64-microcode         install
12 anacron                 install
13 apg                     install
14 app-install-data-partner install
15 apparmor                install
16 appport                 install
17 appport-gtk             install
18 appport-symptoms        install
19 appstream               install
```

b) Find and install package:

i) Query the apt cache and find file manager packages.

```
apt-cache search "file manager"
```

[apt-cache search "query"; search the apt package cache for matching query "file manager"]

```
osboxes@osboxes:~$ apt-cache search "file manager"
gir1.2-nautilus-3.0 - libraries for nautilus components - gir bindings
gnome-control-center - utilities to configure the GNOME desktop
libnautilus-extension-dev - libraries for nautilus components - development ver
sion
libnautilus-extension1a - libraries for nautilus components - runtime version
nautilus - file manager and graphical shell for GNOME
nautilus-data - data files for nautilus
nautilus-sendto - easily send files via email from within Nautilus
nautilus-share - Nautilus extension to share folder using Samba
4pane - four-pane detailed-list file manager
a11y-profile-manager - Accessibility Profile Manager - Command-line utility
a11y-profile-manager-doc - Accessibility Profile Manager - (Documentation
a11y-profile-manager-indicator - Accessibility Profile Manager - Unity desktop
indicator
caja - file manager for the MATE desktop
caja-actions - Caja extension to create custom actions
```

ii) Install and run text based file manager called Midnight Commander.

```
apt-cache search "midnight commander"
```

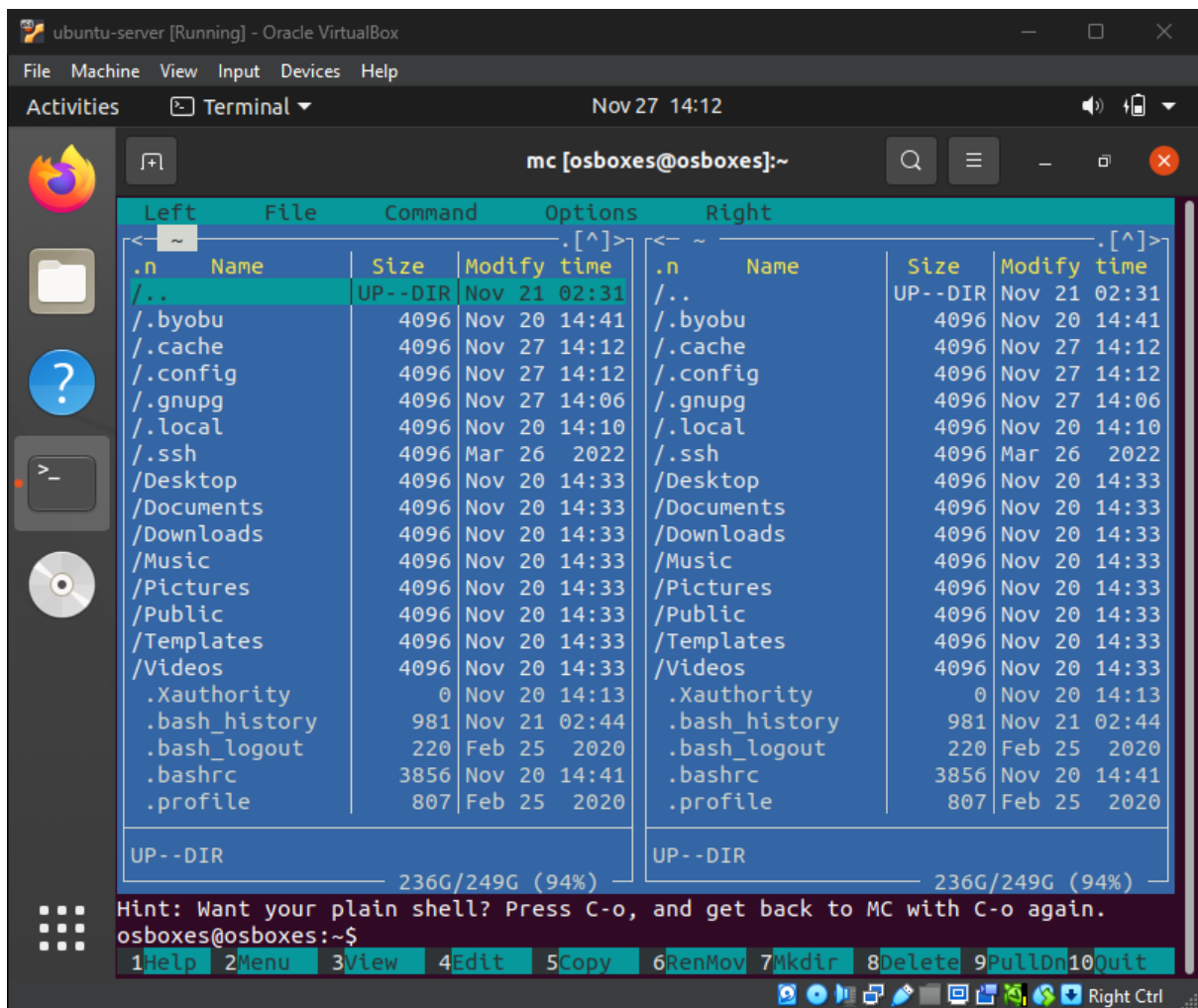
```
osboxes@osboxes:~$ apt-cache search "midnight commander"
avfs - virtual filesystem to access archives, disk images, remote locations
junior-system - Debian Jr. System tools
krusader - twin-panel (commander-style) file manager
mc - Midnight Commander - a powerful file manager
mc-data - Midnight Commander - a powerful file manager -- data files
moc - ncurses based console audio player
pilot - Simple file browser from Alpine, a text-based email client
```

```
sudo apt install mc
```

```
osboxes@osboxes:~$ sudo apt install mc
[sudo] password for osboxes:
Sorry, try again.
[sudo] password for osboxes:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libssh2-1 mc-data
Suggested packages:
  arj catdvi | texlive-binaries dbview djvulibre-bin epub-utils gv
  imagemagick libaspell-dev links | w3m | lynx odt2txt python python-boto
  python-tz
The following NEW packages will be installed:
  libssh2-1 mc mc-data
0 upgraded, 3 newly installed, 0 to remove and 155 not upgraded.
Need to get 1,817 kB of archives.
After this operation, 7,994 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 libssh2-1 a
```

mc

[run Midnight Commander]



iii) Using dpkg -L list all Midnight Commander installed file.

```
dpkg -L mc
```

[lists all Midnight Commander installed files]

```
osboxes@osboxes:~$ dpkg -L mc
/.
/etc
/etc/mc
/etc/mc/edit.indent.rc
/etc/mc/filehighlight.ini
/etc/mc/mc.default.keymap
/etc/mc/mc.emacs.keymap
/etc/mc/mc.ext
/etc/mc/mc.menu
/etc/mc/mcedit.menu
/etc/mc/sfs.ini
/usr
/usr/bin
/usr/bin/mc
/usr/lib
/usr/lib/mc
/usr/lib/mc/cons.saver
/usr/lib/mc/ext.d
/usr/lib/mc/ext.d/archive.sh
/usr/lib/mc/ext.d/doc.sh
/usr/lib/mc/ext.d/image.sh
/usr/lib/mc/ext.d/misc.sh
/usr/lib/mc/ext.d/package.sh
/usr/lib/mc/ext.d/sound.sh
/usr/lib/mc/ext.d/text.sh
/usr/lib/mc/ext.d/video.sh
/usr/lib/mc/ext.d/web.sh
/usr/lib/mc/extfs.d
```

iv) Remove the Midnight Commander packages.

```
sudo apt remove mc
```

[removes package but keeps configuration files]

```
osboxes@osboxes:~$ sudo apt remove mc
[sudo] password for osboxes:
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is h
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is h
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is h
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is h
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is h
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is h
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is h
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is h
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is h
eld by process 6474 (unattended-upgr)... 9s
```

```
sudo apt purge mc
```

[purge completely removes the package, including its configuration files]

```
osboxes@osboxes:~$ sudo apt purge mc
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libssh2-1 mc-data
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  mc*
0 upgraded, 0 newly installed, 1 to remove and 138 not upgraded.
1 not fully installed or removed.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] y
Setting up initramfs-tools (0.136ubuntu6.7) ...
update-initramfs: deferring update (trigger activated)
(Reading database ... 164831 files and directories currently installed.)
Purging configuration files for mc (3:4.8.24-2ubuntu1) ...
Processing triggers for initramfs-tools (0.136ubuntu6.7) ...
update-initramfs: Generating /boot/initrd.img-5.4.0-200-generic
I: The initramfs will attempt to resume from /dev/sda4
I: (UUID=8b9c5f42-b19f-48a1-aed8-fe51c7dcbb59)
I: Set the RESUME variable to override this.

Progress: [ 71%] [#####.....]
```

2. User Management

a) Use ls command to list ALL user in /home directory.

```
ls /home
```

```
osboxes@osboxes:~$ ls /home
eusoff  lost+found  osboxes
osboxes@osboxes:~$
```

b) Using adduser command, add TWO (2) users:

i) Sysadmin – use sysadmin as username and admin888 as password, add sysadmin as sudoer.

```
sudo adduser sysadmin
```

```
osboxes@osboxes:~$ sudo adduser sysadmin
Adding user `sysadmin' ...
Adding new group `sysadmin' (1002) ...
Adding new user `sysadmin' (1002) with group `sysadmin' ...
The home directory `/home/sysadmin' already exists. Not copying from `/etc/skel'.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for sysadmin
Enter the new value, or press ENTER for the default
  Full Name []: sysadmin
    Room Number []: 888
    Work Phone []: 0
    Home Phone []: 0
    Other []: 0
Is the information correct? [Y/n] y
osboxes@osboxes:~$
```

```
sudo usermod -aG sudo sysadmin
```

```
Is the information correct? [Y/n] y
osboxes@osboxes:~$ sudo usermod -aG sudo sysadmin
osboxes@osboxes:~$
```

ii) Linux – use linux as username and linux### as password.

```
sudo adduser linux
```

```
osboxes@osboxes:~$ sudo adduser linux
Adding user `linux' ...
Adding new group `linux' (1003) ...
Adding new user `linux' (1003) with group `linux' ...
Creating home directory `/home/linux' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for linux
Enter the new value, or press ENTER for the default
  Full Name []: linux
    Room Number []: ###
    Work Phone []: 0
    Home Phone []: 0
    Other []: 0
Is the information correct? [Y/n] y
osboxes@osboxes:~$
```

c) Logout from current user and login using sysadmin account.

su sysadmin

```
osboxes@osboxes:~$ su sysadmin
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

sysadmin@osboxes:/home/osboxes$
```

d) Confirm both users and their directory already created:

i) Use ls command in /home directory.

ls /home

```
sysadmin@osboxes:/home/osboxes$ ls /home
eusoff  linux  lost+found  osboxes  sysadmin
```

ii) Using cat or less command, view the user data in /etc/passwd and identify username, userid and groupid field for sysadmin and linux.

cat /etc/passwd | grep -E "sysadmin|linux"

```
sysadmin@osboxes:/home/osboxes$ cat /etc/passwd | grep -E "sysadmin|linux"
sysadmin:x:1002:1002:sysadmin,888,0,0,0:/home/sysadmin:/bin/bash
linux:x:1003:1003:linux,###,0,0,0:/home/linux:/bin/bash
sysadmin@osboxes:/home/osboxes$
```

e) Password Management

i) Using cat or less command, view password data in /etc/shadow.

sudo cat /etc/shadow | grep -E "sysadmin|linux"

```
sysadmin@osboxes:/home/osboxes$ sudo cat /etc/shadow | grep -E "sysadmin|linux"
[sudo] password for sysadmin:
sysadmin:$6$g0mT8pkxY4LQXKSM$z8wblEwsC8tS0NqY73Br18Q/MEI5W6cI9wpGF/K4RezD2Je5qs
.z85Ywx92DwwfbYQ20hCNHjKN0o.wu2GUJU0:20054:0:99999:7:::
linux:$6$vw5Pz9DRpPj7BRMM$JrNue5JJc9Ix/bbTQ5bw5mRLm5eiQMfc8M73hNE2VKclRQ7VvJRjL
OECy2DT/NmtNXvu.fm9WegLIU2P66LXz1:20054:0:99999:7:::
```

ii) Using passwd command, change the linux password to ubuntu***. View the content of /etc/shadow using cat command and identify the changes in password field.

sudo passwd linux

```
sysadmin@osboxes:/home/osboxes$ sudo passwd linux
New password:
Retype new password:
passwd: password updated successfully
sysadmin@osboxes:/home/osboxes$
```

sudo cat /etc/shadow | grep linux

```
sysadmin@osboxes:/home/osboxes$ sudo cat /etc/shadow | grep linux
linux:$6$hzp9jxv9YWrAoCFF$AJkAywnwpIgsId3kjJvAI0ZJEtQD3UXwVglQ.6cd6f0D1cgBgrG/n
kLP2bUP5.ZKaQFiJ7EnzRS0BJFC80FB8.:20054:0:99999:7:::
sysadmin@osboxes:/home/osboxes$
```

3. File and Disk Management

a) Find FOUR (4) different types of files in /dev directory such as symbolic link, block device, character device and directory.

`ls -l /dev`

```
osboxes@osboxes:~$ ls -l /dev
total 0
crw-r--r--  1 root    root      10, 235 Nov 27 14:12 autofs
drwxr-xr-x  2 root    root      360 Nov 27 14:12 block
drwxr-xr-x  2 root    root       80 Nov 27 14:01 bsg
crw-rw----  1 root    disk     10, 234 Nov 27 14:12 btrfs-control
drwxr-xr-x  3 root    root       60 Nov 27 14:01 bus
lrwxrwxrwx  1 root    root        3 Nov 27 14:12 cdrom -> sr0
drwxr-xr-x  2 root    root    3760 Nov 27 14:16 char
crw--w----  1 root    tty       5,   1 Nov 27 14:12 console
lrwxrwxrwx  1 root    root      11 Nov 27 14:01 core -> /proc/kcore
drwxr-xr-x  4 root    root       80 Nov 27 14:01 cpu
crw-----  1 root    root     10,  59 Nov 27 14:12 cpu_dma_latency
crw-----  1 root    root     10, 203 Nov 27 14:01 cuse
drwxr-xr-x  7 root    root      140 Nov 27 14:01 disk
drwxr-xr-x  3 root    root      100 Nov 27 14:01 dri
lrwxrwxrwx  1 root    root        3 Nov 27 14:12 dvd -> sr0
```

symbolic link: l

`ls -l /dev | grep -E "^[l]"`

```
osboxes@osboxes:~$ ls -l /dev | grep -E "^[l]"
lrwxrwxrwx  1 root    root        3 Nov 28 02:14 cdrom -> sr0
lrwxrwxrwx  1 root    root      11 Nov 28 02:14 core -> /proc/kcore
lrwxrwxrwx  1 root    root        3 Nov 28 02:14 dvd -> sr0
lrwxrwxrwx  1 root    root      13 Nov 28 02:14 fd -> /proc/self/fd
lrwxrwxrwx  1 root    root      12 Nov 28 02:14 initctl -> /run/initctl
lrwxrwxrwx  1 root    root      28 Nov 28 02:14 log -> /run/systemd/journal
/dev-log
```

block device: b

`ls -l /dev | grep -E "^[b]"`

```
osboxes@osboxes:~$ ls -l /dev | grep -E "^[b]"
brw-rw----  1 root    disk       7,   0 Nov 28 02:14 loop0
brw-rw----  1 root    disk       7,   1 Nov 28 02:14 loop1
brw-rw----  1 root    disk       7,   2 Nov 28 02:14 loop2
brw-rw----  1 root    disk       7,   3 Nov 28 02:14 loop3
brw-rw----  1 root    disk       7,   4 Nov 28 02:14 loop4
brw-rw----  1 root    disk       7,   5 Nov 28 02:14 loop5
brw-rw----  1 root    disk       7,   6 Nov 28 02:14 loop6
```

character device: c

`ls -l /dev | grep -E "^[c]"`

```
osboxes@osboxes:~$ ls -l /dev | grep -E "^[c]"
crw-r--r--  1 root    root     10, 235 Nov 28 02:14 autofs
crw-rw----  1 root    disk     10, 234 Nov 28 02:14 btrfs-control
crw--w----  1 root    tty       5,   1 Nov 28 02:15 console
crw-----  1 root    root     10,  60 Nov 28 02:14 cpu_dma_latency
crw-----  1 root    root     10, 203 Nov 28 02:14 cuse
crw-----  1 root    root     10,  62 Nov 28 02:14 ecryptfs
```

directory: d

`ls -l /dev | grep -E "^[d]"`

```

Help
osboxes@osboxes:~$ ls -l /dev | grep -E "^[d]"
drwxr-xr-x  2 root    root      360 Nov 28 02:15 block
drwxr-xr-x  2 root    root       80 Nov 28 02:14 bsg
drwxr-xr-x  3 root    root       60 Nov 28 02:14 bus
drwxr-xr-x  2 root    root    3720 Nov 28 02:14 char
drwxr-xr-x  4 root    root       80 Nov 28 02:14 cpu
drwxr-xr-x  7 root    root      140 Nov 28 02:14 disk

```

```
ls -l /dev | grep -E "^[bcdl]"
```

```

osboxes@osboxes:~$ ls -l /dev | grep -E "^[bcdl]"
crw-r--r--  1 root    root      10, 235 Nov 27 14:12 autofs
drwxr-xr-x  2 root    root      360 Nov 27 14:12 block
drwxr-xr-x  2 root    root       80 Nov 27 14:01 bsg
crw-rw----  1 root    disk     10, 234 Nov 27 14:12 btrfs-control
drwxr-xr-x  3 root    root       60 Nov 27 14:01 bus
lrwxrwxrwx  1 root    root        3 Nov 27 14:12 cdrom -> sr0
drwxr-xr-x  2 root    root    3760 Nov 27 14:16 char
crw--w----  1 root    tty       5,   1 Nov 27 14:12 console
lrwxrwxrwx  1 root    root      11 Nov 27 14:01 core -> /proc/kcore
drwxr-xr-x  4 root    root       80 Nov 27 14:01 cpu
crw-----  1 root    root     10,  59 Nov 27 14:12 cpu_dma_latency
crw-----  1 root    root     10, 203 Nov 27 14:01 cuse
drwxr-xr-x  7 root    root      140 Nov 27 14:01 disk
drwxr-xr-x  3 root    root     100 Nov 27 14:01 dri
lrwxrwxrwx  1 root    root        3 Nov 27 14:12 dvd -> sr0
crw-----  1 root    root     10,  62 Nov 27 14:12 ecryptfs
crw-rw----  1 root    video    29,   0 Nov 27 14:12 fb0
lrwxrwxrwx  1 root    root        3 Nov 27 14:01 fd -> /proc/self/fd

```

b) Create directory called Car.

```
mkdir car
```

```
osboxes@osboxes:~$ mkdir car
```

c) Create a file called Proton using touch command. Give others the write permission to the file.

```
touch Proton
```

```
chmod o+w Proton
```

```

osboxes@osboxes:~$ touch Proton
osboxes@osboxes:~$ chmod o+w Proton
osboxes@osboxes:~$

```

d) Edit the file using nano, add Saga, Waja, Iswara, X90 and X70 as its content.

```
nano Proton
```

```
osboxes@osboxes:~$ nano Proton
```

Insert car name inside as content.

```

GNU nano 4.8
Saga
Waja
Iswara
X90
X70

```

save by pressing Ctrl + O, then Enter to save.

Exit by pressing Ctrl + X.

e) Rename Proton to Models and move the file to Car directory.

```
mv Proton Models
```

[I wrote wrong 😊 in command hehe]

```
osboxes@osboxes:~$ mv Proton Car
osboxes@osboxes:~$ mv Car Models
```

```
mv Models Car
```

```
osboxes@osboxes:~$ mv Models Car
osboxes@osboxes:~$
```

f) Delete Car directory and its contents.

```
rm -r Car
```

```
osboxes@osboxes:~$ rm -r Car
```

g) List all block devices.

```
lsblk
```

[lists block devices in tree format, showing device names, types, and mount points]

```
osboxes@osboxes:~$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
loop0       7:0      0  73.9M  1 loop /snap/core22/1663
loop1       7:1      0  61.9M  1 loop /snap/core20/1328
loop2       7:2      0  91.9M  1 loop /snap/lxd/29619
loop3       7:3      0  67.2M  1 loop /snap/lxd/21835
loop4       7:4      0  38.8M  1 loop /snap/snapd/21759
loop5       7:5      0  63.7M  1 loop /snap/core20/2434
loop6       7:6      0   1.5G  1 loop /snap/root-framework/936
loop7       7:7      0  44.3M  1 loop /snap/snapd/23258
sda         8:0      0   500G  0 disk
├─sda1      8:1      0    1M  0 part
├─sda2      8:2      0   236G  0 part /
├─sda3      8:3      0   300M  0 part /boot
├─sda4      8:4      0    9.5G  0 part [SWAP]
└─sda5      8:5      0  254.2G  0 part /home
sr0        11:0     1    57M  0 rom  /media/osboxes/VBox_GAs_7.1.4
```

h) Report files system disk space usage.

```
df -h
```

[-h displays sizes in human readable format (GB,MB etc)

```
osboxes@osboxes:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G   0    1.9G   0% /dev
tmpfs           394M  1.5M  392M   1% /run
/dev/sda2       232G  11G  209G   5% /
tmpfs           2.0G   0    2.0G   0% /dev/shm
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
tmpfs           2.0G   0    2.0G   0% /sys/fs/cgroup
/dev/loop2      92M   92M    0 100% /snap/lxd/29619
/dev/loop1      62M   62M    0 100% /snap/core20/1328
/dev/loop0      74M   74M    0 100% /snap/core22/1663
/dev/loop3      68M   68M    0 100% /snap/lxd/21835
/dev/loop4      39M   39M    0 100% /snap/snapd/21759
/dev/loop5      64M   64M    0 100% /snap/core20/2434
/dev/loop6     1.6G  1.6G    0 100% /snap/root-framework/936
/dev/sda5       250G  207M  237G   1% /home
/dev/sda3       275M  215M   40M  85% /boot
tmpfs           394M  28K  394M   1% /run/user/1000
/dev/sr0        57M   57M    0 100% /media/osboxes/VBox_GAs_7.1.4
/dev/loop7      45M   45M    0 100% /snap/snapd/23258
```